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Workshop

Purpose

The `workshop/` directory contains material intended to support demonstrations, guided exercises, experiments, or training activities using the OT security testbed.

Learning Goals

A workshop based on this repository can introduce participants to:

1. OT/ICS architecture
2. PLC and SCADA interaction
3. Modbus/TCP communication
4. Normal industrial traffic
5. Packet capture and protocol analysis
6. Controlled security testing
7. The relationship between network activity and physical/process state

Suggested Exercise Flow

1. Start the Environment

Launch the Docker-based OT services and verify that the containers are healthy.

2. Inspect the PLC

Use OpenPLC to understand the simulated control logic and available process variables.

3. Observe SCADA

Open ScadaBR and confirm that the operator interface reflects PLC state.

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4. Generate Normal Traffic

Run the normal traffic generator and capture the resulting Modbus/TCP packets.

5. Analyze Packets

Use Wireshark to identify Modbus transactions, function codes, addresses, and values.

6. Perform a Controlled Test

Use the dedicated laboratory testing components to perform a reversible Modbus write operation.

7. Observe the Effect

Check the PLC state, SCADA display, and network capture.

8. Document Findings

Record the request, response, affected data point, observed process change, and relevant packet fields.

Workshop Safety

All exercises should remain inside the isolated test environment. Security-testing activities should be authorized, controlled, reversible, and limited to the simulated OT assets.

Related Components

- `openplc/`
- `scadabr/`
- `traffic/`
- `modbus-client/`
- `attacker/`

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